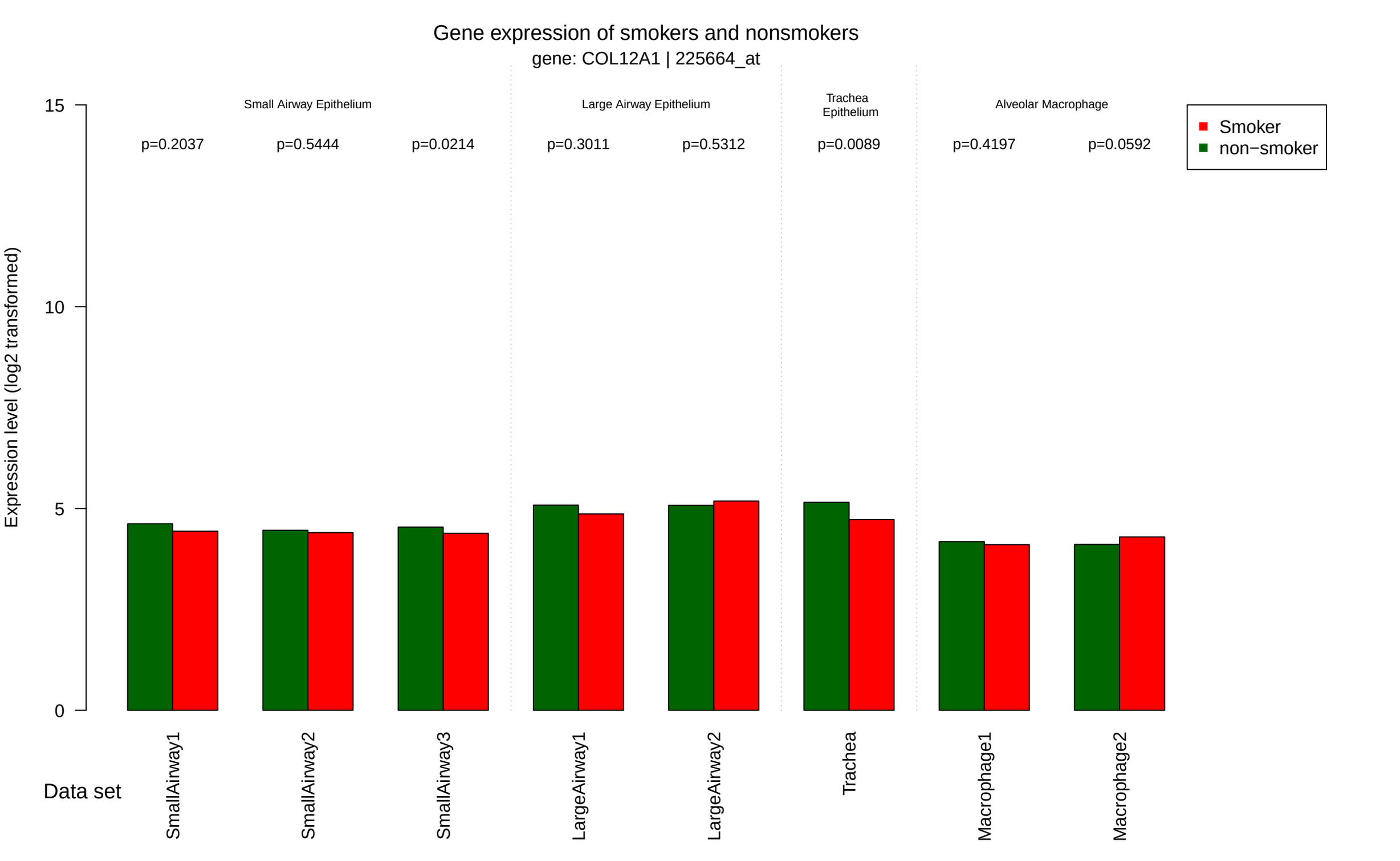
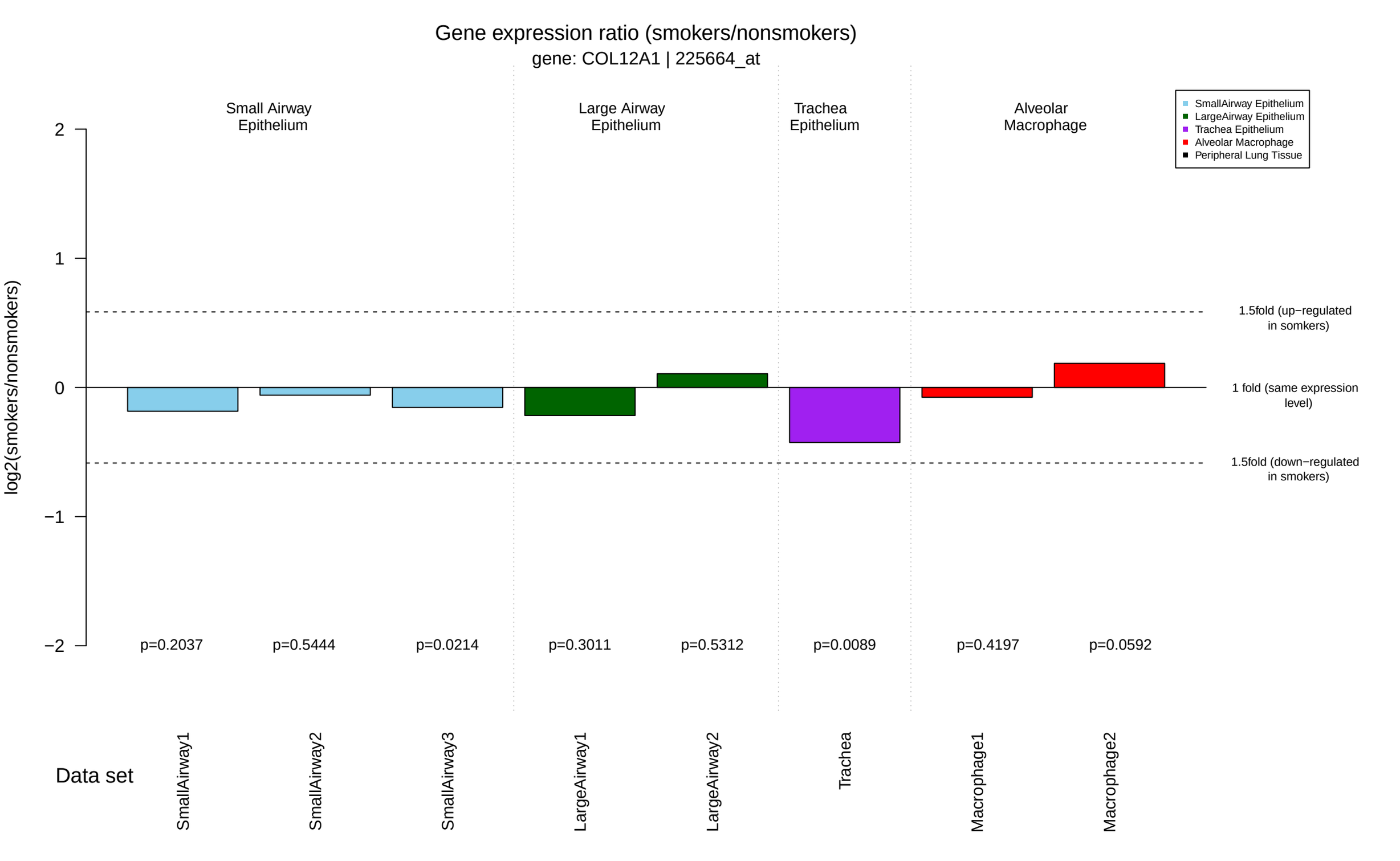
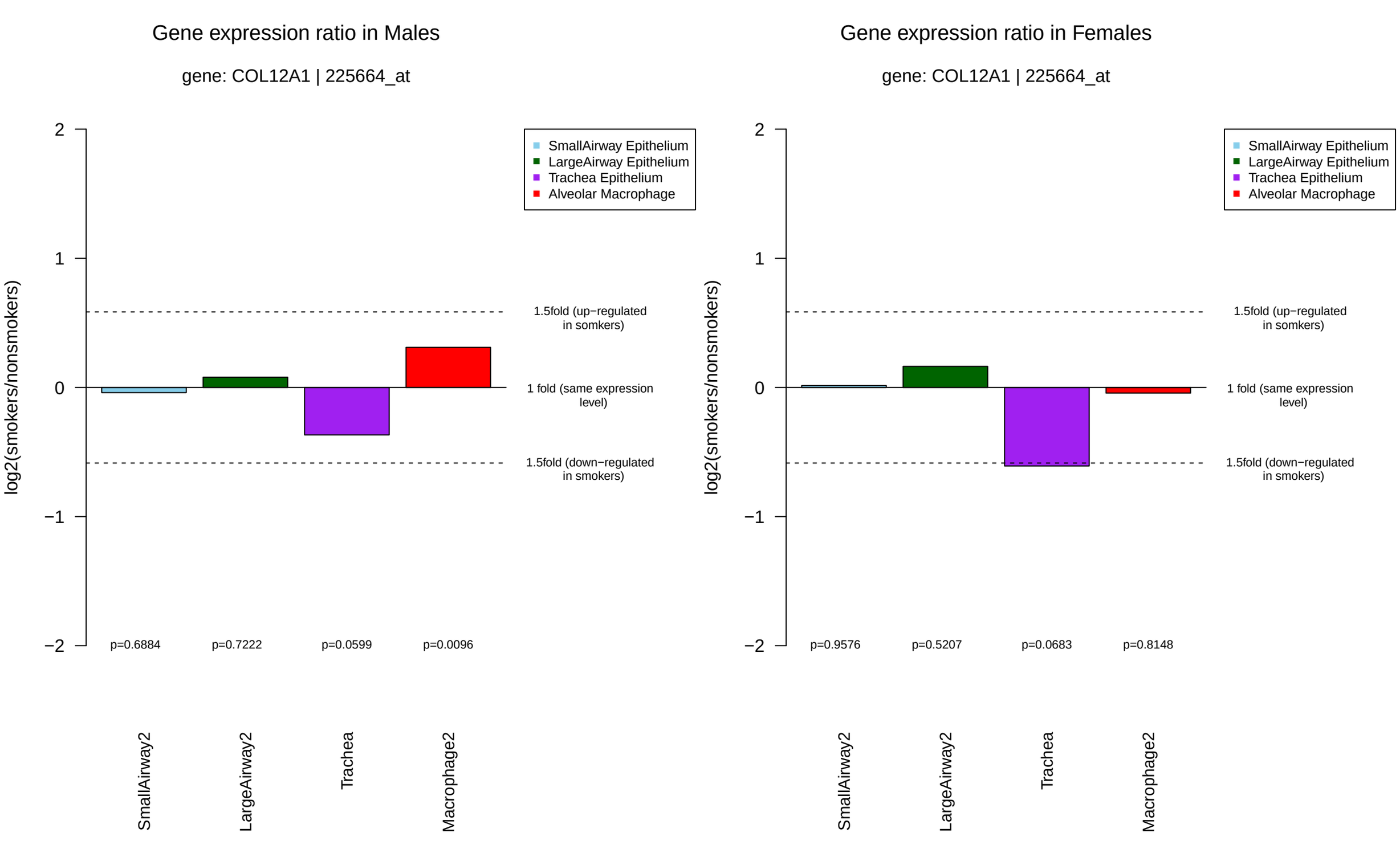
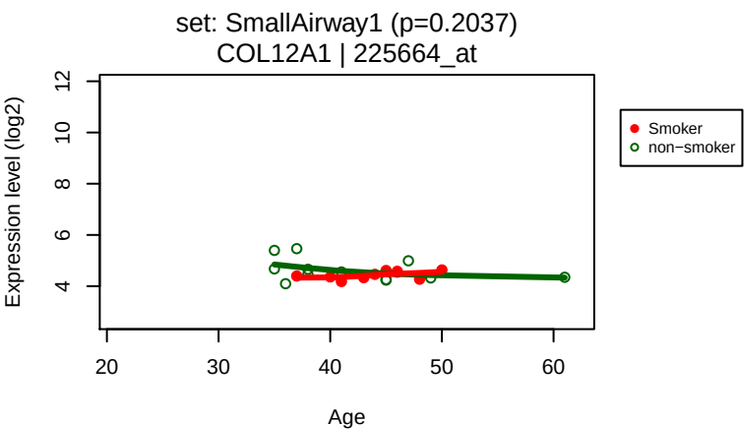
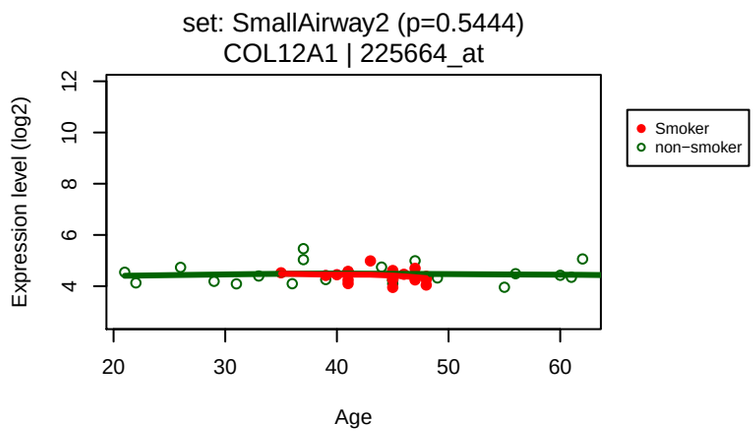
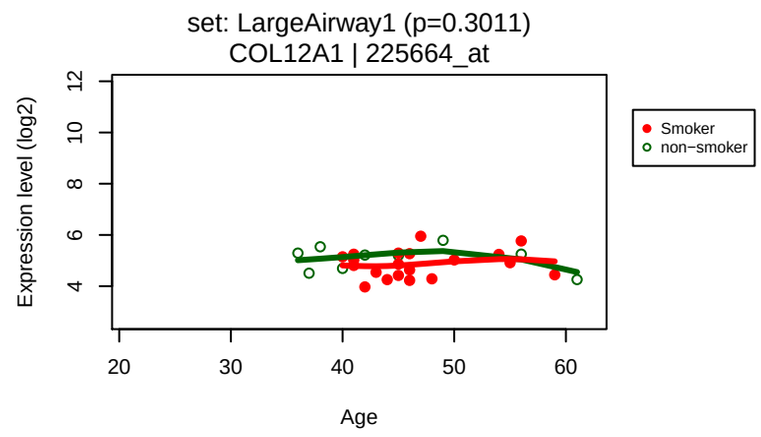

Gene expression of smokers and nonsmokers
gene: COL12A1 | 225664_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2037
p=0.5444
p=0.0214
p=0.3011
p=0.5312
p=0.0089
p=0.4197
p=0.0592
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

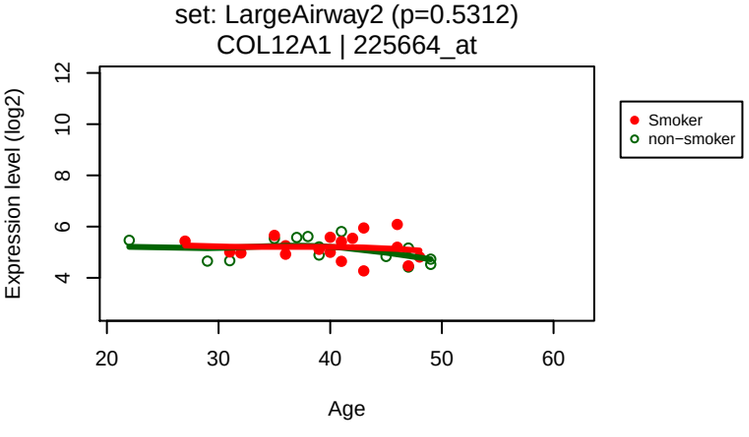

Gene expression ratio (smokers/nonsmokers)
gene: COL12A1 | 225664_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2037
p=0.5444
p=0.0214
p=0.3011
p=0.5312
p=0.0089
p=0.4197
p=0.0592
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

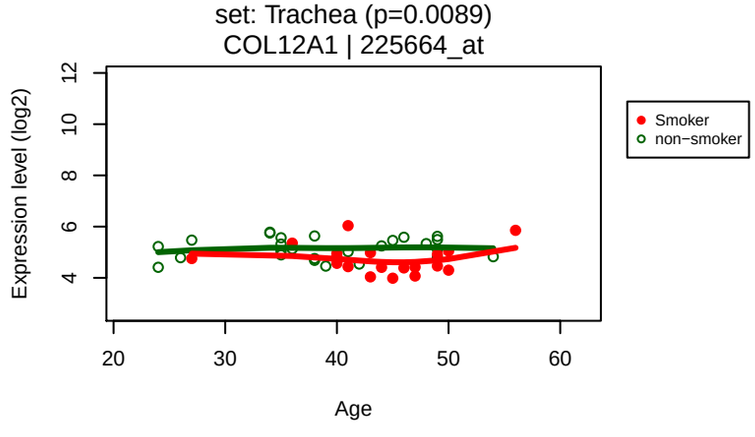

Gene expression ratio in Males
gene: COL12A1 | 225664_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6884
p=0.7222
p=0.0599
p=0.0096
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL12A1 | 225664_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9576
p=0.5207
p=0.0683
p=0.8148
SmallAirway2
LargeAirway2
Trachea
Macrophage2

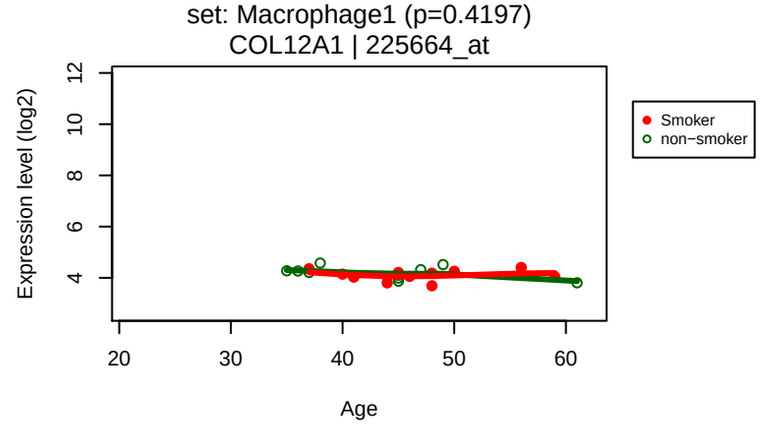

set: SmallAirway1 (p=0.2037)
COL12A1 | 225664_at
Expression level (log2)
Age
Smoker
non−smoker

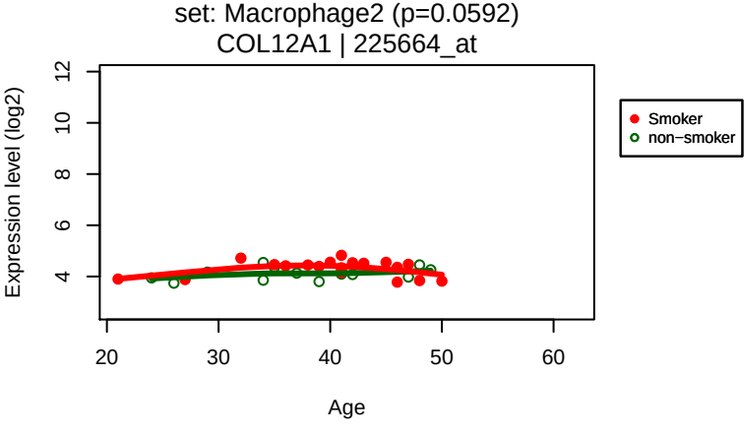

set: SmallAirway2 (p=0.5444)
COL12A1 | 225664_at
Expression level (log2)
Age
Smoker
non−smoker

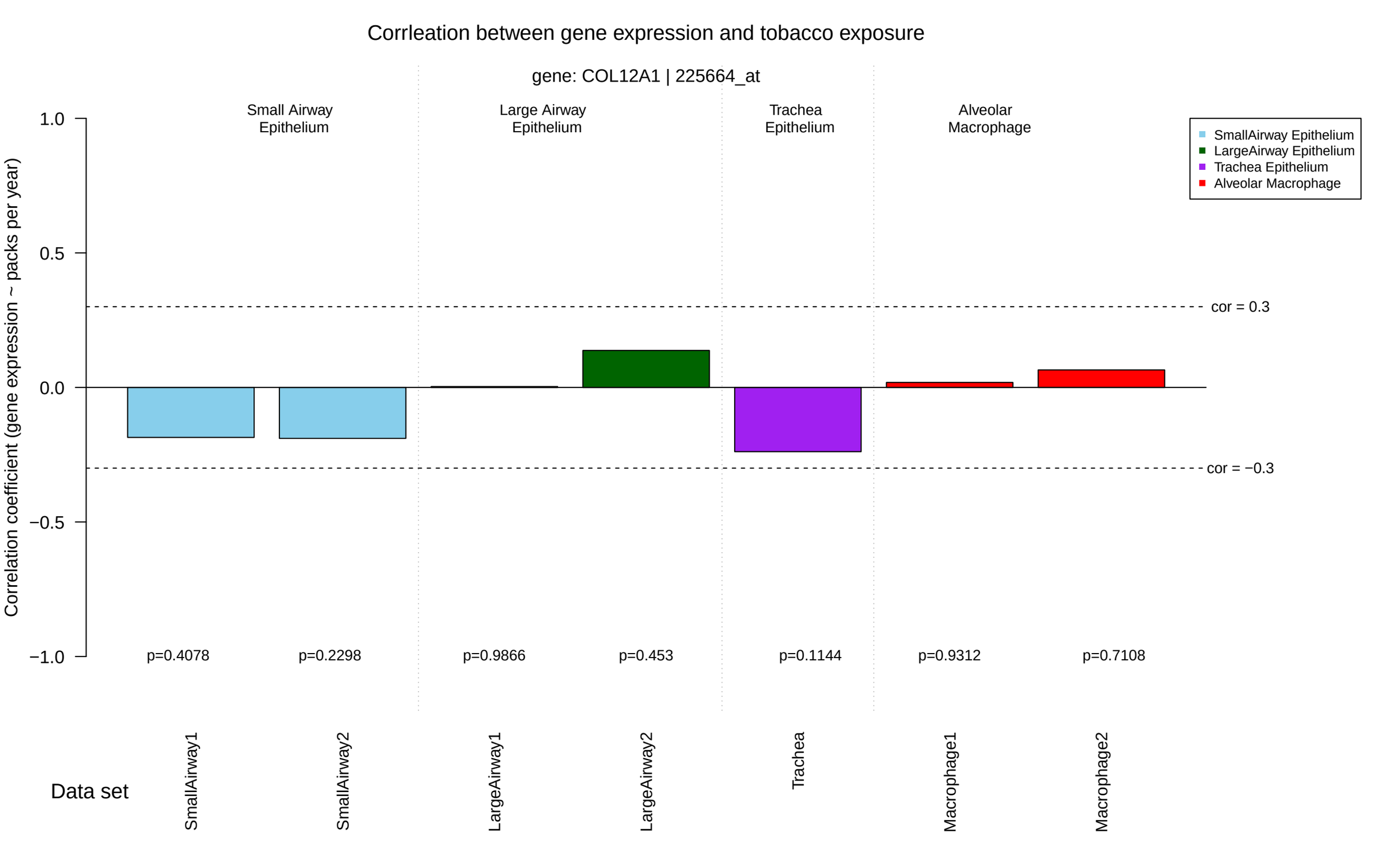

set: LargeAirway1 (p=0.3011)
COL12A1 | 225664_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5312)
COL12A1 | 225664_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0089)
COL12A1 | 225664_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4197)
COL12A1 | 225664_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0592)
COL12A1 | 225664_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL12A1 | 225664_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4078
p=0.2298
p=0.9866
p=0.453
p=0.1144
p=0.9312
p=0.7108
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2